GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

Challenge Name: Harnessing the Generative Quantum Eigensolver for Next-Generation Materials Design				
Phase #: 2				
Team Name: Entangled Trio				
First Name	**Last Name**	**Email**	**Aqora Username**	**Role within Team**
Niraj	Venkat	venkatn93@gmail.com	nirajvenkat	Project Manager / Team Coordinator
Mackenson	Polché	polchemackenson@yahoo.com.mx	mack_polche	Domain Expert
Muskan	Aman	www.muskanaman@gmail.com	muscaanmnmnm	Data / Technical Lead

1 Focus Area, Rationale, and Stakeholder Relevance

This project implements a dual extreme ultraviolet (EUV) simulation pipeline to model both the photoabsorption and subsequent core-level Auger electron decay in photoresist materials to evaluate their chemical sensitivity and performance. During EUV lithography, 92 eV exposure first induces photoabsorption to excite or ionize core electrons (e.g., Iodine 4d shell), which subsequently undergo Auger decay cascades. These secondary electrons initiate chemical reactions such as photoacid generation which create patterns in the resists. Understanding both the initial absorption and the subsequent electron emission is essential for optimizing resist sensitivity and resolution at sub-10 nm nodes. The interactive simulation code for our EUV simulation pipeline has been published as a pre-rendered Jupyter notebook [1].

The primary stakeholders and beneficiaries of this work include:

- **Semiconductor Manufacturers and Lithographers:** Guides the predictive design of photoresists to maximize light-absorption and secondary electron yield.
- **Materials Scientists:** Provides a robust computational framework for predicting electron cascade kinetics in halogenated compounds.

2 Technical Methodology and Hybrid Quantum Integration

Our EUV simulation pipeline is written in Pennylane [2], and models both photoabsorption and Auger decay:

1. **Photoabsorption Spectrum:** Using a time-domain Green's function propagated via a second-order Trotter product formula on a Compressed Double-Factorized (CDF) Hamiltonian, based on [3].
2. **Auger Electron KE Spectrum:** Using a Generative Quantum Eigensolver (GQE) [4] combined with the Quantum Self-Consistent Equation-of-Motion (q-sc-EOM) and One-Center Approximation (OCA), based on [5].

2.1 EUV Photoabsorption Spectrum Simulation

The EUV photoabsorption channel simulates the dipole-allowed valence-excitation spectrum under 92 eV radiation. We compute the transition dipole moment operators $\hat{m}_\rho$ in the molecular active space and project the ground state wavefunction to obtain initial states $\hat{m}_\rho|I\rangle$ along the Cartesian x, y, z coordinates. We then propagate the states in the time-domain after which measuring the time-dependent expectation values $\langle\hat{m}_\rho(t)\rangle$ yields the time-domain Green's function $G(t)$, which is Fourier-transformed to resolve the absorption spectrum in the frequency domain.

2.2 EUV Auger Emission: The GQE Challenge and Decay Modeling

Simulating the Auger emission spectrum requires modeling multi-electron electronic decay from core-ionized and doubly-ionized states. Since the decay transitions are highly sensitive to the ground-state reference, preparing a high-fidelity reference state represents a major GQE challenge. We address this using a hybrid quantum workflow consisting of three stages.

2.2.1 GQE Ground-State Reference Preparation

To prepare high-fidelity reference ground states, we employ a GQE that generates shallow quantum circuits. We implement two transformer architectures in PyTorch:

1. **GPTQE:** An autoregressive causal model based on nanoGPT that predicts discrete gate sequences.
2. **DiTQE:** A Diffusion Transformer-style architecture where the target ground-state energy acts as a continuous condition. DiTQE uses multi-headed self-attention with adaptive RMSNorm to modulate token embeddings, capturing global circuit topology and eliminating directional biases.

2.2.2 Excited-State Extraction via q-sc-EOM

Using the optimized GQE reference state $|0\rangle = U_{\mathrm{GQE}}|\mathrm{HF}\rangle$ as the vacuum, we extract excited states via the q-sc-EOM solver [6] across two channels:

- **Ionization Potential (IP) Space:** Resolves core-ionized ($N-1$ electrons) doublet states $|\mathrm{IP}_I\rangle$.
- **Double Ionization Potential (DIP) Space:** Resolves doubly-ionized ($N-2$ electrons) singlet states $|\mathrm{DIP}_K\rangle$.

We compute the transition reduced density matrix (RDM) $R_{KI;\alpha\beta} = \langle \mathrm{DIP}_K|\hat{a}_c^\dagger \hat{a}_\beta \hat{a}_\alpha|\mathrm{IP}_I\rangle$ involving one-electron creation and two-electron annihilation.

2.2.3 Minimal Basis Projection and One-Center Approximation (OCA)

To computationally simulate the Auger decay of H_2O:

- **MBS Localization:** MO coefficients C in the primitive CGTO basis are localized onto the core-hole emitter site via the projection $D = T^{-1}UC$. The OCA restricts the final two-electron Coulomb elements strictly to this single atomic center, neglecting multi-center continuum wave distributions.
- **Three-Body Transition RDM:** Evaluates localized decay amplitudes via $B_{KI;lm} = \sum_{\alpha,\beta} R_{KI;\alpha\beta} V_{lm\alpha\beta}$.
- **Spectral Resolution:** Computes partial decay rates via Fermi's Golden Rule ($\Gamma_{KI;lm} = 2\pi|B_{KI;lm}|^2$), determines kinetic energies $E_{\mathrm{kin}} = E_{\mathrm{IP}}^{(I)} - E_{\mathrm{DIP}}^{(K)}$, and we resolve the continuous spectrum via Lorentzian broadening at 1.5 eV.

3 Data Modeling and Benchmarking Strategy

Our data modeling strategy utilizes the standard STO-3G basis set. To scale both the photoabsorption and the core-level emission calculations, we incorporate three acceleration techniques:

1. **Active Space Selection (AVAS):** Trims the molecular orbitals down to active valence electrons and orbitals (e.g., CAS(24e, 18o) for 36 qubits or CAS(16e, 16o) for 32 qubits in IMePh) to maintain classical tractability.
2. **Compressed Double Factorization (CDF):** Compresses the molecular Hamiltonian terms using a Block-Invariant Symmetry Shift (BLISS) and L2-regularized numerical fitting, minimizing the one-norm of the two-body operator.
3. **Direct OpenMolcas HDF5 Integration:** Rather than using proxy molecular integrals, we integrate direct multi-center atomic integrals extracted from OpenMolcas HDF5 files for accurate OCA transition rate modeling.

For this project, we opted to run our pipeline on the water (H_2O) molecule (7 spatial orbitals, 14 qubits) to reproduce published results in the quantum chemistry literature. Our classical benchmarks include exact FCI from PySCF [7] to validate GQE training, and RASSI in OpenMolcas [8] to validate the Auger spectrum simulation.

4 Experimental Results and Discussion

For the **absorption** simulation, spectrum in the 10–100 eV range shows good agreement with the exact classical CASCI baseline, resolving the physical valence excitations in the 10–45 eV regime and dropping to a flat zero baseline in the 45–100 eV range. This confirms that H_2O has no dipole-allowed transitions near the 92 eV EUV regime in a minimal STO-3G basis, which is expected. This serves as a key stepping stone for photoresist-adjacent molecules, e.g., IMePh (4-iodo-2-methylphenol, C_7H_7IO) where Iodine 4d core-excitations are active at 92 eV.

For the **emission** simulation, GQE reference state preparation is the core challenge. The GQE training was evaluated over 10,000 epochs. GQE energy statistics compared against random sequences are summarized in Table 1.

Table 1: GQE Ground State Energy (Hartrees) of H_2O

Source	**Average Energy**	**Minimum Energy**	**Maximum Energy**	**Error to Ground State**
GPT	-74.9533	-74.9650	-74.9131	0.0475
DiT	-74.9327	-74.9715	-74.7391	0.0409
FCI	-75.0125	–	–	0.0000

The GPT model successfully optimized the excitation sequences, lowering the minimum energy to -74.9650 Hartrees. DiTQE was used to evaluate the benefits of energy-conditioned generative transformer architectures against the GPTQE baseline. We identified that the initial poor performance of the DiTQE model relative to the baseline was caused by a causal leakage bug in the multi-headed self-attention (MHA) block. Because the attention mechanism did not originally apply a lower-triangular causal mask, the model was able to look ahead at future tokens during autoregressive training. Correcting this by enforcing causal masking in the MHA block resolved this look-ahead data leak, aligning training attention with autoregressive sequential generation and enabling DiTQE to learn sequence representations without cheating.

Using GQE reference states, q-sc-EOM resolved the excited states, while MBS and OCA yielded Auger decay rates. We successfully reproduce the main peak at $E_{\text{kin}} = 502.4$ eV, which aligns with [5] and experimental results [9].

5 Quantum Platform Justification and Resource Needs

Requested: 8xB200-180GB-SXM6 (Inferior alternative: 8xH100-80GB-SXM5).
FP64 state-vector memory: 35 qubits (549.7 GB), 36 qubits (1,099.5 GB), 37 qubits (2,199.0 GB).
Our rationale for selecting Blackwell (B200) over H100:

- **Latency**: Halves the GPU count for NVLink distribution vs H100 (2 vs 4 GPUs for 34 qubits); SXM6 NVLink-5 bandwidth (1.8 TB/s) minimizes inter-GPU communication bottleneck.
- **Capacity**: 8xB200 offers 1,440 GB VRAM, fitting the 36-qubit space (1,099.5 GB) on a single node. 8xH100 is capped at 640 GB and will go out of memory.
- **Precision**: SXM6 native FP64 throughput prevents rounding errors in the kinetic energy distributions.

6 Gap Analysis and Implementation Roadmap

Our Phase 3 roadmap focuses on scaling to 36 qubits and targeting large molecules like IMePh. We will apply our AVAS and CDF implementations to scale GQE execution. We plan to incorporate pseudopotentials to replace the core electrons of heavy elements like Iodine. Lastly, we will maximize the throughput of our pipeline. By leveraging CUDAQ, we can ensure that our pipeline is GPU-accelerated and hardware-native.

References

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